Technical Document #648: Cloud Computing - Comprehensive Overview

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Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Kubernetes orchestrates containerized applications across clusters of machines.
- Containerization packages applications with their dependencies.
- Platform as a Service (PaaS) provides a platform for developing and deploying applications.